



Qualification Pack

Course in Web Technology

Backend Development: Programming with PHP/ Backend Development:
Programming with Python

QP Code: HAR/ITS/Q1701

Version: 1.0

NSQF Level: 4

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Qualification Pack

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HAR/ITS/Q1701: Course in Web Technology

Brief Job Description

The course imparts practical and technical knowledge to its learners about various client side and server side web technologies that are used to create web applications.

Personal Attributes

Creative, detail-oriented, and analytical professional with strong problem-solving, communication, teamwork, and adaptability skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HAR/ITS/N0910: IT Foundation and Digital Tools \(CWT\)](#)
2. [HAR/ITS/N1703: Database Management with SQL](#)
3. [HAR/ITS/N1702: Frontend Development: Building Responsive Websites](#)
4. [HAR/ITS/N1701: UX/UI Design with Figma and Web Application Frameworks](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
6. [HAR/ITS/N0802: OJT/ Project & Viva \(1\)](#)

Electives (mandatory to select at least one):

Elective 1: Backend Development: Programming with PHP

1. [HAR/ITS/N1704: Programming with PHP](#)

Elective 2: Backend Development: Programming with Python

1. [HAR/ITS/N1705: Programming with Python \(CWT\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
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Sub-Sector	IT- Services
Occupation	Web Developer
Country	India
NSQF Level	4
Credits	20
Aligned to NCO/ISCO/ISIC Code	2513.0101
Minimum Educational Qualification & Experience	12th grade Pass OR 12th Class (pursuing) OR 10th grade pass plus 2-year NTC OR 10th grade pass with 1 year NTC plus NAC OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-04-IT-03526-2025-V2-HARTRON
NQR Version	2

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HAR/ITS/N0910: IT Foundation and Digital Tools (CWT)

Description

The NOS provides basic knowledge and skills to operate computer and prepares you to perform basic office jobs.

Scope

The scope covers the following :

- The NOS builds essential skills in Word, Excel, PowerPoint, and Internet usage, preparing students for basic office tasks and serving as a foundation for advanced computer applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the meaning, characteristics, history, generations, and classification of computers.
- PC2.** Identify and explain the main components, input/output devices, and the working of a computer system.
- PC3.** Differentiate between data and information, and perform basic number conversions.
- PC4.** Demonstrate understanding of operating systems, including the settings and functions of Windows..
- PC5.** Diagnose and resolve common peripheral malfunctions and connectivity issues.
- PC6.** Create and edit documents using word processing tools with proper formatting and layout.
- PC7.** Apply formulas and functions in spreadsheets and use spreadsheet tools effectively.
- PC8.** Design and present information visually using digital presentation software.
- PC9.** Develop correct typing posture and accuracy with improved speed.
- PC10.** Utilize networking and internet skills to communicate online, manage email accounts, ensure cybersecurity, and understand cloud storage

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Demonstrates understanding of the evolution, characteristics, and classification of computers, including their historical development and technological generations.
- KU2.** Identifies and explains the main hardware components, input/output devices, and the overall functional process of a computer system.
- KU3.** Explains the purpose, functions, and configuration of operating systems, particularly Windows, in managing computer resources and settings.
- KU4.** Understands methods for diagnosing and resolving common hardware, peripheral, and connectivity issues within a computer system.



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- KU5.** Demonstrates knowledge of digital productivity tools, including word processors, spreadsheets, and presentation software, for creating and managing professional documents and visual materials.
- KU6.** Distinguishes between data and information and applies basic number system conversions relevant to computing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** set goals for career development.
- GS2.** efficiently plan daily data entry tasks to meet deadlines.
- GS3.** Identify and resolve data inconsistencies or system errors.
- GS4.** Ensure accuracy and confidentiality of customer data.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	31	15	30	-
PC1. Understand the meaning, characteristics, history, generations, and classification of computers.	-	-	-	-
PC2. Identify and explain the main components, input/output devices, and the working of a computer system.	-	-	-	-
PC3. Differentiate between data and information, and perform basic number conversions.	-	-	-	-
PC4. Demonstrate understanding of operating systems, including the settings and functions of Windows..	-	-	-	-
PC5. Diagnose and resolve common peripheral malfunctions and connectivity issues.	-	-	-	-
PC6. Create and edit documents using word processing tools with proper formatting and layout.	-	-	-	-
PC7. Apply formulas and functions in spreadsheets and use spreadsheet tools effectively.	-	-	-	-
PC8. Design and present information visually using digital presentation software.	-	-	-	-
PC9. Develop correct typing posture and accuracy with improved speed.	-	-	-	-
PC10. Utilize networking and internet skills to communicate online, manage email accounts, ensure cybersecurity, and understand cloud storage	-	-	-	-
NOS Total	31	15	30	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0910
NOS Name	IT Foundation and Digital Tools (CWT)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Data Entry Operator
NSQF Level	2.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HAR/ITS/N1703: Database Management with SQL

Description

Understand the concepts of database and SQL queries

Scope

The scope covers the following :

- Create and maintain database

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain database concepts, DBMS architecture, relational models, ER models, normalization, and different types of database keys.
- PC2.** Create, modify, and manage databases and tables using SQL commands including DDL, DML, TCL, DCL, and DQL operations.
- PC3.** Perform data retrieval and manipulation using SQL queries, joins, set operations, views, subqueries, triggers, and stored procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The requirement and objective of organization and scope of work
- KU2.** Basic concepts of RDBMS and its structure
- KU3.** Designing and organizing data into database efficiently
- KU4.** Creating queries effectively for data retrieval and manipulation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prioritise work based on the requirement.
- GS2.** Produce work output as per the requirement in a timely manner.
- GS3.** Listen actively and communicate with other team members in a simple and effective terms.
- GS4.** Seek inputs from clients and other team members to create a clear and effective database design

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	15	8	-
PC1. Explain database concepts, DBMS architecture, relational models, ER models, normalization, and different types of database keys.	-	-	-	-
PC2. Create, modify, and manage databases and tables using SQL commands including DDL, DML, TCL, DCL, and DQL operations.	-	-	-	-
PC3. Perform data retrieval and manipulation using SQL queries, joins, set operations, views, subqueries, triggers, and stored procedures.	-	-	-	-
NOS Total	9	15	8	-



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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1703
NOS Name	Database Management with SQL
Sector	Information Technology Sector
Sub-Sector	
Occupation	Web Developer
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HAR/ITS/N1702: Frontend Development: Building Responsive Websites

Description

This NOS provides knowledge and practical skills in frontend web development using HTML, CSS, JavaScript, jQuery, and XML for creating interactive and responsive websites.

Scope

The scope covers the following :

- The scope includes designing and developing responsive web pages using HTML5, CSS2/CSS3, JavaScript, Bootstrap, and jQuery. It also covers XML concepts, document validation, DOM manipulation, form handling, media integration, responsive layouts, animations, and interactive web functionalities for modern web applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Create and format web pages using HTML, CSS, and HTML5 elements including forms, multimedia, tables, and responsive design features.
- PC2.** Develop responsive web pages using viewport settings, grid systems, media queries, images, videos, and Bootstrap framework.
- PC3.** Apply CSS styling techniques and JavaScript programming concepts to develop interactive and dynamic web pages.
- PC4.** Use XML, DTD, XML Schema, XPath, and XQuery concepts for structuring, validating, and querying XML documents.
- PC5.** Implement JavaScript concepts including variables, functions, loops, events, BOM, DOM manipulation, and form validation in web applications.
- PC6.** Use jQuery syntax, selectors, events, effects, traversing, and AJAX features to create interactive web interfaces.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the concepts, principles, tools, and technologies used in frontend web development including HTML, CSS, JavaScript, jQuery, XML, and responsive web design.
- KU2.** webpage structuring, styling techniques, client-side scripting, DOM manipulation, XML validation, responsive layouts, and interactive web application development.
- KU3.** the role of frontend web development in website creation, digital communication, and user experience enhancement across industries.
- KU4.** to apply web development standards, coding practices, responsive design principles, and quality assurance procedures followed in software and web development environments.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Demonstrate effective oral and written communication skills for discussing website requirements, design concepts, and technical documentation.
- GS2.** Apply analytical thinking and problem-solving skills while developing, testing, and debugging responsive web pages and scripts.
- GS3.** Plan and organize web development tasks, digital assets, and project activities efficiently while meeting quality standards and deadlines.
- GS4.** Exhibit professionalism, teamwork, adaptability, customer-centric approach, and attention to detail while working in web development environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	36	15	10	-
PC1. Create and format web pages using HTML, CSS, and HTML5 elements including forms, multimedia, tables, and responsive design features.	-	-	-	-
PC2. Develop responsive web pages using viewport settings, grid systems, media queries, images, videos, and Bootstrap framework.	-	-	-	-
PC3. Apply CSS styling techniques and JavaScript programming concepts to develop interactive and dynamic web pages.	-	-	-	-
PC4. Use XML, DTD, XML Schema, XPath, and XQuery concepts for structuring, validating, and querying XML documents.	-	-	-	-
PC5. Implement JavaScript concepts including variables, functions, loops, events, BOM, DOM manipulation, and form validation in web applications.	-	-	-	-
PC6. Use jQuery syntax, selectors, events, effects, traversing, and AJAX features to create interactive web interfaces.	-	-	-	-
NOS Total	36	15	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1702
NOS Name	Frontend Development: Building Responsive Websites
Sector	Information Technology Sector
Sub-Sector	
Occupation	Web Developer
NSQF Level	4
Credits	3.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HAR/ITS/N1701: UX/UI Design with Figma and Web Application Frameworks

Description

This NOS provides knowledge and practical understanding of UX/UI design concepts using Figma and the fundamentals of web application frameworks.

Scope

The scope covers the following :

- The scope includes designing user interfaces and prototypes using Figma tools and understanding UX/UI principles for enhancing user experience.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the concepts, similarities, and differences between User Experience (UX) and User Interface (UI) design.
- PC2.** Use Figma tools and interface features to create wireframes, layouts, and high-fidelity mock-ups for web and mobile applications.
- PC3.** Apply UX/UI design principles to develop user-friendly, visually appealing, and interactive interface designs.
- PC4.** Explain the purpose, structure, and types of web application frameworks used in frontend and backend development.
- PC5.** Differentiate between Content Management Systems (CMS) and web application frameworks based on functionality and usage.
- PC6.** Describe MVC architecture and its workflow involving Model, View, and Controller components in application development.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the concepts, principles, tools, and techniques of UX/UI design, wireframing, prototyping, and high-fidelity mock-up creation using Figma.
- KU2.** web application frameworks, MVC architecture, frontend and backend frameworks, and their role in developing modern web applications.
- KU3.** the importance of UX/UI design and web application frameworks in enhancing user experience, application performance, and digital business solutions.
- KU4.** to apply design standards, development practices, collaboration methods, and workflow processes followed in web design and software development environments.

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** Demonstrate effective oral and written communication skills for presenting UX/UI concepts, design ideas, and framework-related documentation.
- GS2.** Plan and organize design workflows, prototypes, development tasks, and project activities efficiently while meeting quality standards and timelines.
- GS3.** Exhibit professionalism, teamwork, creativity, adaptability, customer-centric approach, and attention to detail while working in UX/UI and web development environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	-	2	-
PC1. Explain the concepts, similarities, and differences between User Experience (UX) and User Interface (UI) design.	-	-	-	-
PC2. Use Figma tools and interface features to create wireframes, layouts, and high-fidelity mock-ups for web and mobile applications.	-	-	-	-
PC3. Apply UX/UI design principles to develop user-friendly, visually appealing, and interactive interface designs.	-	-	-	-
PC4. Explain the purpose, structure, and types of web application frameworks used in frontend and backend development.	-	-	-	-
PC5. Differentiate between Content Management Systems (CMS) and web application frameworks based on functionality and usage.	-	-	-	-
PC6. Describe MVC architecture and its workflow involving Model, View, and Controller components in application development.	-	-	-	-
NOS Total	9	-	2	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1701
NOS Name	UX/UI Design with Figma and Web Application Frameworks
Sector	Information Technology Sector
Sub-Sector	
Occupation	Web Developer
NSQF Level	4
Credits	1.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

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HAR/ITS/N0802: OJT/ Project & Viva (1)

Description

The NOS enabling students to apply their acquired skills and knowledge in real work or job-oriented environments.

Scope

The scope covers the following :

- Provides practical exposure, enhance skills, and prepare students for real work environments by bridging the gap between theory and practice

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrates application of theoretical knowledge in practical tasks

PC2. Adapts to workplace requirements and follows instructions

PC3. Demonstrates clear understanding of concepts during viva

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Workplace practices and professional standards

KU2. Basic problem-solving techniques

KU3. Communication skills in a work environment

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Work collaboratively as part of a team

GS2. Follow instructions and organizational procedures

GS3. Manage time and meet deadline

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	-	30	30
PC1. Demonstrates application of theoretical knowledge in practical tasks	-	-	-	-
PC2. Adapts to workplace requirements and follows instructions	-	-	-	-
PC3. Demonstrates clear understanding of concepts during viva	-	-	-	-
NOS Total	-	-	30	30

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0802
NOS Name	OJT/ Project & Viva (1)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Programmer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HAR/ITS/N1704: Programming with PHP

Description

Learn PHP Programming

Scope

The scope covers the following :

- Create and maintain dynamic websites.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Know how to install and use WAMP / XAMPP server.
- PC2.** Know and use PHP syntax, conditional structure, arrays and functions.
- PC3.** Able to use POST and GET methods
- PC4.** Understand the concept of cookies and sessions
- PC5.** Know to connect and interact with database.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The requirement & objective of organization and scope of work.
- KU2.** Basic syntax, structures and OOPS concept.
- KU3.** Create and manage database in phpMyAdmin
- KU4.** PHP's role in server side scripting
- KU5.** Handling Form data and session management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Produce work output as per the predefined format and in a timely manner.
- GS2.** Listen actively and communicate with other team members in simple and effective terms.
- GS3.** Seek inputs from clients and other team members for clear and impressive design.
- GS4.** Prioritise work based on urgency
- GS5.** Ability to breakdown and solve the problem efficiently.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	24	15	10	-
PC1. Know how to install and use WAMP / XAMPP server.	-	-	-	-
PC2. Know and use PHP syntax, conditional structure, arrays and functions.	-	-	-	-
PC3. Able to use POST and GET methods	-	-	-	-
PC4. Understand the concept of cookies and sessions	-	-	-	-
PC5. Know to connect and interact with database.	-	-	-	-
NOS Total	24	15	10	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1704
NOS Name	Programming with PHP
Sector	Information Technology Sector
Sub-Sector	
Occupation	Web Developer
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Qualification Pack

HAR/ITS/N1705: Programming with Python (CWT)

Description

The NOS introduces learners to the fundamentals and advanced concepts of Python programming. It covers Python environment setup, syntax, variables, operators, control statements, data structures, functions, file handling, modules, object-oriented programming, exception handling, regular expressions, and database connectivity with MySQL.

Scope

The scope covers the following :

- Provide learners with comprehensive knowledge and practical skills in Python programming, covering fundamentals, control structures, data handling, functions, file management, modules, object-oriented programming, exception handling, regular expressions, and database connectivity with MySQL.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Install and configure Python environment and develop programs using correct syntax, variables, operators, and data types.
- PC2.** Apply control statements, loops, functions, and data structures to solve programming problems efficiently.
- PC3.** Perform file handling, module creation, package management, and exception handling in Python applications.
- PC4.** Implement object-oriented programming concepts such as classes, inheritance, encapsulation, and method overriding.
- PC5.** Develop database-driven applications using Python and MySQL with CRUD operations and regular expression validation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of Python programming including syntax, variables, data types, operators, control statements, and data structures.
- KU2.** Concepts of functions, modules, packages, object-oriented programming, exception handling, and regular expressions in Python.
- KU3.** Techniques for file handling, database connectivity, MySQL operations, and development of database-driven applications.
- KU4.** Software development practices, debugging methods, coding standards, and safe handling of program data and files.
- KU5.** Organizational requirements related to documentation, teamwork, problem solving, data security, and professional coding practices.



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read, understand, and prepare technical documents, program codes, and project reports effectively.
- GS2.** Communicate clearly with team members, trainers, and clients regarding technical requirements and solutions.
- GS3.** Apply analytical thinking and problem-solving skills for debugging and developing efficient Python applications.
- GS4.** Demonstrate planning, organizing, and time management skills while completing programming tasks and projects.
- GS5.** Follow professional ethics, teamwork, customer-centric approach, and decision-making practices in software development activities.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	24	15	10	-
PC1. Install and configure Python environment and develop programs using correct syntax, variables, operators, and data types.	-	-	-	-
PC2. Apply control statements, loops, functions, and data structures to solve programming problems efficiently.	-	-	-	-
PC3. Perform file handling, module creation, package management, and exception handling in Python applications.	-	-	-	-
PC4. Implement object-oriented programming concepts such as classes, inheritance, encapsulation, and method overriding.	-	-	-	-
PC5. Develop database-driven applications using Python and MySQL with CRUD operations and regular expression validation.	-	-	-	-
NOS Total	24	15	10	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1705
NOS Name	Programming with Python (CWT)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Web Developer
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment is done online after completion of the Course and is integrated into the Qualification Pack (QP), of which the NOS is a part.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N0910.IT Foundation and Digital Tools (CWT)	31	15	30	-	76	25
HAR/ITS/N1703.Database Management with SQL	9	15	8	-	32	11
HAR/ITS/N1702.Frontend Development: Building Responsive Websites	36	15	10	-	61	20
HAR/ITS/N1701.UX/UI Design with Figma and Web Application Frameworks	9	-	2	-	11	4
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	4
HAR/ITS/N0802.OJT/ Project & Viva (1)	-	-	30	30	60	20
Total	105	75	80	30	290	84

Elective: 1 Backend Development: Programming with PHP

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N1704.Programming with PHP	24	15	10	-	49	16
Total	24	15	10	-	49	16

Elective: 2 Backend Development: Programming with Python

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N1705.Programming with Python (CWT)	24	15	10	-	49	16
Total	24	15	10	-	49	16



Qualification Pack

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.